Md Obaidul Haque

PhD Candidate
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RESEARCH INTERESTS

Applied and Computational Mathematics; Mathematical Biology; Infectious Disease Modeling; Networked Agent-Based Simulations; Integrating Generative AI and LLMs into Epidemic Dynamics; Applied Stochastic Processes.

ACADEMIC APPOINTMENTS

2025–Present **Graduate Research Assistant**, Department of Mathematics, The University of Alabama, Tuscaloosa, AL.
Supervisor: Dr. Jia Zhao

2022–2025 **Graduate Teaching Assistant**, Department of Mathematics, The University of Alabama, Tuscaloosa, AL.

2020–2022 **Lecturer in Mathematics**, Department of Mathematics, Hamdard University Bangladesh, Munshiganj, Bangladesh.

EDUCATION

Expected 2027 **Ph.D. in Mathematics**, The University of Alabama, Tuscaloosa, AL.
Advisor: Dr. Jia Zhao
Focus: Hybrid Agent-Based Modeling, Networked Epidemiology, and Applied Stochastic Processes.

2025 **M.A. in Mathematics**, The University of Alabama, Tuscaloosa, AL.

2019 **M.Sc. in Mathematics**, Comilla University, Cumilla, Bangladesh.

2018 **B.Sc. in Mathematics**, Comilla University, Cumilla, Bangladesh.

PUBLICATIONS

Manuscripts Under Review (1)

[R1] J. Wan, **M. O. Haque**, K. Ho, J. Giuffre, C. Cheng, and J. Zhao (2026). A hybrid networked SEIR model with generative-AI-driven agents for behavioral heterogeneity in epidemic modeling. *Manuscript under review*.

Manuscripts in Preparation (2)

[P2] A. Gurung, **M. O. Haque**, C. Wang, and J. Zhao. A multi-agent stochastic SEIRS compartmental framework with independent Poisson clocks and distilled LLM-based behavioral decision rules. *In preparation*, target submission November 2026.

[P1] **M. O. Haque**†, J. Lee, C. Wang, and J. Zhao. A martingale formulation for agent-based SIR models with fixed protective behaviour: finite-size effects at the node and pair approximations. *In preparation*, target submission Spring 2027.

PRESENTATIONS

Minisymposium and Presentations

2026 **Haque, M. O.** "Recent Developments and Emerging Directions in Mathematical Biology." *Minisymposium Presentation*, 11th SIAM Central States Section Annual Meeting, Jackson State University, Jackson, MS.

2026 **Haque, M. O.** "LLM-Driven Behavioral Heterogeneity in Networked SEIR(S) Models with GAM Distillation." *Presentation*, 38th Annual University of Alabama System Applied Mathematics Meeting, University of Alabama, Tuscaloosa, AL.

FELLOWSHIPS AND AWARDS

2026 **NSF Travel Award**, 11th SIAM Central States Section Annual Meeting.

2026 **Graduate Travel Grant**, Department of Mathematics, The University of Alabama.

2018–2019 **National Science and Technology (NST) Fellowship**, Ministry of Science and Technology, Bangladesh.

2018–2019 **Teacher-Student Research Fellowship**, Comilla University, Bangladesh.

2018 **Runner-Up**, 1st Intra-University Mathematics Olympiad, Comilla University, Bangladesh.

2015, 2016 **Participant**, 6th & 7th National Undergraduate Mathematics Olympiad, Bangladesh.

TEACHING EXPERIENCE

Instructor of Record

Department of Mathematics, The University of Alabama

Calculus and Its Applications (MATH 121)	Summer 2025
Applied Differential Equations I (MATH 238 · two sections)	Spring 2025
Calculus III (MATH 227)	Fall 2024
Calculus I (MATH 125)	Spring 2024
Departmental Tutoring & Grading (Math Teaching and Learning Centre)	Fall 2022–Fall 2023

Department of Mathematics, Hamdard University Bangladesh

Calculus II (MATH122)	Spring 2022
Differential Equations (MATH224)	Spring 2022
Linear Algebra (MATH223)	Fall 2021
Discrete Mathematics (MATH213)	Fall 2021
Introductory Numerical Analysis (MATH312)	Spring 2021
Calculus I (MATH112)	Spring 2021

Teaching Development

2025–Present **Teaching Excellence Across the Curriculum in Higher Education (TEACHE)**, Barefield College of Arts & Sciences, The University of Alabama
Structured doctoral teaching certification; encompasses peer observations, course design, assessment matrix, online teaching course, and full teaching portfolio.

COMPUTATIONAL SKILLS

Languages	Python, MATLAB, C/C++, FORTRAN
Tools	LaTeX, Overleaf, Git/GitHub, Linux/Unix, Jupyter
Methods	CTMC, Agent-Based Modeling (ABM), ODE/PDE solvers

PROFESSIONAL SERVICE

Book Author: *Shongkhyar Ananda* (Joy of Numbers), an introductory mathematics book in the Bengali language exploring interesting numbers and patterns.

Publication Editor: *BigganKothon*, a student science magazine, Comilla University, Bangladesh.

Leadership: Former President, Comilla University Science Club, Bangladesh; Former General Secretary, Brahmanbaria Science Club, Bangladesh.

PROFESSIONAL MEMBERSHIPS

AMS	American Mathematical Society
SIAM	Society for Industrial and Applied Mathematics
APS	American Physical Society
AWM	Association for Women in Mathematics (UA Chapter)
COMAP	Consortium for Mathematics and Its Applications

REFERENCES

Dr. Jia Zhao
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Department of Mathematics
The University of Alabama, Tuscaloosa, AL 35487
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